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Robust Libor Modelling and Pricing of Derivative Products

John Schoenmakers



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Preface

This book may be considered, on the one hand, as a fast and handy introduction to the area of (Libor) interest rate modelling and pricing, intended for students in finance, junior financial engineers, and other interested people with a mathematical background, which covers some knowledge of basic probability theory and stochastic processes.¹ On the other hand, due to an innovative treatment of issues concerning Libor calibration and pricing of exotic instruments, including new valuation procedures for products with Bermudan style exercise features, the book is suited particularly to the more experienced practitioner and the researcher in this field as well.

Since its development back in the mid nineties the Libor Market Model is still one of the most popular and advanced tools for modelling interest rates and interest rate derivatives at the so called fixed income desks. Nevertheless, a useful procedure for “implied” calibration of the model, which avoids historical or subjective information but exclusively relies on market prices of liquidly traded products, has been a perennial problem. In recent studies, however, we² tackled this problem and developed fast and robust implied methods for calibrating Libor. This calibration approach, its impact on Libor derivative pricing, and the development of new methods for pricing complicated exotic (Bermudan) structures in the Libor model provided the principle motivation for writing this book.

In fact, several key ideas and results presented in this work were obtained during our research and consulting on Libor modelling and derivative pricing over the last years. Practically right after the development and appearance of the Libor market model in early versions of the now broadly known standard papers of Brace, Gatarek & Musiela, Jamshidian, and Miltersen, Sandman & Sonderman, we got involved in consulting projects on implementation and application of this new promising model for pricing exotic interest rate products at fixed income desks. Naturally, the first job was to construct stable methods for calibration of the model to liquidly traded caps and swaptions. However, soon it turned out that in spite of many nice properties of the Libor market model, calibration of a more factor model (i.e., more than one driving random source) tends to suffer from stability problems. In particular, difficulties of intrinsic nature came out by the determination of the trade off between the explanation power of forward Libor volatilities versus forward Libor correlations from market data, even if these quantities were stuck into a very parsimonious framework. Loosely speaking, even in a parsimonious setting it turned out possible to match the same market quotes with essentially different calibrations! In the present work we present a remedy for this problem via incorporation of a collateral swaption market criterion in the cal-

¹The reader is recommended to consult some standard texts on this subject if necessary.

²Throughout the text “we” stands for the present author and his circle of research associates.

ibration procedure. Basically, via this collateral market concept stability is retained in the calibration. Indeed, for a parsimonious model the behaviour of the model parameters implied by the market via a thus enhanced calibration procedure proved to be very regular when going from one calibration date to another, similar to implied Black-Scholes volatilities observed from standard stock options!

In the area of Bermudan style exotic Libor products we expose effective standard methods such as the Andersen method for constructing lower bounds and the Rogers method for constructing (dual) upper bounds. Moreover, we introduce a recently developed generic iteration procedure (Kolodko & Schoenmakers 2004, Bender & Schoenmakers 2004) for the valuation of standard Bermudan and multiple callable Bermudan instruments. A very appealing feature of the presented iteration procedure is the fact that it applies to any underlying model because one only needs to simulate the underlying process just like in the (naïve) backward dynamic program. In contrast to the backward dynamic program, however, the degree of nesting of simulations does not grow with the number of exercise dates, but can be chosen in advance. Usually, a quadratic Monte Carlo simulation gives extremely good results.

The robust implied calibration approach has been implemented together with pricing routines for various exotic products such as trigger swaps, flexi caps and Bermudan swaptions. Using this software, realised as an Excel add-in, we illustrate in various practical cases the calibration procedure and its impact on exotics pricing.

The material is embedded in a compact and self contained review of the necessary financial mathematical theory on interest rate modelling given in Chapter 1.

Chapter 2 concentrates on economically sensible parametrisations of the Libor Market Model. Particularly, the presented low parameter correlation structures in this chapter may be considered as new.

In Chapter 3 we expose an intrinsic stability problem connected with direct least squares calibration methods, supported by both practical examples, and examples generated in our “laboratory”. We here introduce a stable calibration procedure by incorporating a criterion related to the swaption market.

Chapter 4 deals with Libor exotics pricing based on real life calibrated models and we here consider the impact of different calibrations on a variety of popular exotic derivative structures. For instance, we show that certain instruments (e.g., the ratchet cap) are quite sensitive for the number of driving factors in the model.

Our new developments on Bermudan style derivative pricing are presented in Chapter 5. In this chapter we also explain the Andersen (1999) method, dual upper bound methods due to Rogers, Haugh & Kogan, and Jamshidian, and introduce an effective algorithm for constructing dual upper bounds.

In Chapter 6 we deal with suitable lognormal approximations for the Libor model and show that in comparison with standard Euler simulation of the Li-

bor dynamics, it is more efficient to simulate prices for long dated instruments using such approximations.

We think it is fair to say that the thorny stability issues concerning (implied) calibration of Libor models, and its consequences for derivative pricing, are not well emphasized and discussed in the literature so far. Moreover, particularly on the subject of Bermudan style derivatives, several new ideas which may serve as a starting point for further research are presented.

Throughout the book I (the author) strove for clarity at every point of issue, and supporting examples and illustrations are given where possible. A trial version of the Excel add-in, used for the empirical experiments, is available on request.

Acknowledgments. This book was composed at the Weierstrass Institute for Applied Analysis and Stochastics in Berlin and I (the author) thank the institute, and particularly my direct colleagues, for the pleasant working environment. Much of the new material presented in this book is developed in cooperation with the DFG research center MATHEON, *Mathematics for key technologies*, in Berlin which deserves my thanks as well. A special thanks goes to my former colleague Oliver Reiß for valuable discussions and developing together with me the C++/Excel prototype implementation. Further I would like to thank, at the Weierstrass Institute, Christian Bender and Anastasia Kolodko for many useful remarks. This work also strongly benefited from cooperation with many persons outside WIAS, among others I mention Brian Coffey, Arnold Heemink, Hermann Haaf, Farshid Jamshidian, Jörg Kienitz, Pieter Klaassen, Peter Kloeden, Christoph März, Grigori Milstein, Geert Jan Olsder, Thorsten Sauder, Samuel Schwalm, and Martin Schweizer.

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Symbol Description

$\emptyset$	the empty set	$\mathbb{R}_{++}$	the set $\{x \in \mathbb{R} : x > 0\}$
$A \cup B$	the union of sets A and B	$\mathbb{R}^m$	the m -dimensional Euclidean space; $\mathbb{R}^1 = \mathbb{R}$
$A \cap B$	the intersection of sets A and B	$:=$	defined as or denoted by
$a \in A$	a belongs to the set A	$\equiv$	identically equal to
$a \notin A$	a belongs not to the set A	$\approx$	approximately equal to
$\mathbb{N}$	The set $\{1,2,3,\dots\}$	$\sim$	distributed as
$\mathbb{N}_0$	The set $\{0,1,2,3,\dots\}$	(a, b)	the open interval $a < x < b$ in $\mathbb{R}$
$\mathbb{R}$	the set of real numbers		
$\mathbb{R}_+$	the set $\{x \in \mathbb{R} : x \geq 0\}$	$[a, b]$	the closed interval

$a \leq x \leq b$ in $\mathbb{R}$	$1_A(x)$	for a set A equals 1 when $x \in A$, 0 when $x \notin A$
$\max(a, b)$ maximum of two real numbers a and b	1_P	for a boolean P equals 1 when P is true, 0 when P is false
$\min(a, b)$ minimum of two real numbers a and b	$A^\top$	the transpose of the matrix (or vector) A
$\inf A$ the infimum of a set A	$ x $	norm $\sqrt{x^\top x}$ when $x \in \mathbb{R}^m$, $m \geq 1$.
$\sup A$ the supremum of a set A	$\text{span}\{v_1, \dots, v_k\}$	the set of linear combinations $\sum_{i=1}^k \alpha_i v_i$ of elements v_i of a linear space
$O(t^q)$, $t \rightarrow 0$ or ∞ expression divided by t^q is bounded for $t \rightarrow 0$ or ∞ , resp.	$\text{rank}(A)$	the dimension of the column space of a matrix A
$o(t^q)$, $t \rightarrow 0$ or ∞ expression divided by t^q converges to zero for $t \rightarrow 0$ or ∞ , resp.	EX	the mathematical expectation of a random variable X
$a \vee b$ logical a "or" b for booleans a and b , $\max(a, b)$ for real numbers a and b	$E^\mathcal{F} X$	the conditional expectation X given the σ -algebra $\mathcal{F}$
$a \wedge b$ logical a "and" b for booleans a and b , $\min(a, b)$ for real numbers a and b	a.s.	almost surely
a^+ $\max(a, 0)$	Var	Variance
a^- $-\min(a, 0)$	Cov	Covariance
	Cor	Correlation